

Utkarsh Singh

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Professional Summary

CS undergraduate (9.3+ CGPA) specializing in Applied AI, RAG pipelines, and deep learning. Proven track record of shipping end-to-end ML applications, from a multilingual clinical chatbot to an exoplanet detector.

Education

Bachelor of Technology, Computer Science and Business Systems	July 2023 – June 2027 (expected)
Bharati Vidyapeeth Deemed University, Pune	CGPA: 9.36
Higher Secondary (Class XII) – PCM & Psychology	April 2015 – March 2023
City Montessori School, Lucknow	XII: 94% X: 91.8%

Skills

Languages: Python, SQL, JavaScript, TypeScript

AI/ML: Deep Learning (CNNs), NLP, LLMs, Computer Vision, Explainable AI(SHAP), RAG Architecture, Feature Engineering, Model Optimization, Ensemble Learning, SMOTE

Web & Databases: React, Next.js, Node.js, Fast API, Chroma DB, PostgreSQL, Supabase, Prisma ORM, REST APIs

Frameworks & Tools: TensorFlow, Scikit-learn, Git, GitHub, VS Code, Streamlit Cloud, SentenceTransformers, Hugging Face

Certifications: OpenAI GPT-4, Intro to Data Science, Information Security, NLP: Infosys Springboard; Python Programming: Coursera

Projects

Sinter (AI Perspective & Counter-Argument Engine) | Github May 2026
Node.js, React, PostgreSQL (Supabase), Llama-3, RAG Architecture

- Architected a **RAG** Chrome extension fetching **5** sources/query, deduplicating into **8** unique snippets; replaced hallucinated LLM citations with live **Serper API** retrieval, guaranteeing **100%** verifiable source URLs.
- Engineered a narrative heuristic engine evaluating political framing across a **100-point** bias spectrum via **Llama-3.3-70b** (0.2 temperature), synthesizing exactly **3** counter-arguments per article from a **5,000-character** context window.
- Designed a **PostgreSQL** “Steel-man vs. Straw-man” crowdsourced arbitration system with a caching layer — reducing query latency from **6s** to **<100ms** for repeat article analyses.

exHUMA: AI Exoplanet Detector | Deployment link January 2026
TensorFlow, CNNs, SHAP, SMOTE

- Built a **CNN-MLP hybrid** scanning **5,087 stars**, flagging **111 candidates** and verifying **5 exoplanets** at **100% recall** overcoming the accuracy paradox where a naive model hit 99% accuracy at **0% recall**.
- Eliminated **95%** of false alarms via Phase Folding and **SNR ≥ 1.5** threshold; applied **SMOTE** with custom class weighting to handle **<1%** positive sample rate.
- Integrated **SHAP** with a custom reshaping wrapper for transit-dip attention heatmaps; reduced manual vetting time by **80%** vs. baseline human review.

Manas Mitra: Multilingual Student Support Chatbot | Collaborative Project | Deployment Link December 2025

Next.js, FastAPI, Gemini API, RAG (ChromaDB), SentenceTransformers

- Architected a clinical **RAG pipeline** using **ChromaDB** and local multilingual-e5-small embeddings to map user queries to **25 distinct CBT frameworks**, ensuring clinically-aligned, hallucination-free generation via **Gemini 2.0**.
- Engineered a zero-latency **Next.js** frontend circuit breaker and a 3-model LLM fallback chain, bypassing API rate-limits and instantly routing acute distress queries to national helplines.

Leadership & Achievements

- Spearheaded ‘imPROMPTu’ at Bharatiyam 2026 as **CSBS Deputy Joint Secretary**; Advanced to **Round 2 of HackWithInfy 2026**; served as **AI/ML Associate** for Google Developer Student Clubs (**GDSC**).